RAHUL MITRA

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SUMMARY

AI and data engineer with industry experience at Abbott and STMicroelectronics, computer vision research on an Amazon Robotics brief, and a civic platform created in collaboration with multiple public sector entities, including People's Association, PA ICT (GovTech), and Tengah Community. Final-year NUS Industrial and Systems Engineering and Computer Science student.

EDUCATION

National University of Singapore **Aug 2023 – Jul 2027**
Bachelor of Engineering, Industrial and Systems Engineering (GPA: 4.37/5.00)

- Pursuing a Second Major in Computer Science and a Minor in Mathematics.
- Awarded Certificate of Outstanding Performance in 3D Computer Vision (CS4277): top student in a class of 24, NUS School of Computing.
- Relevant coursework: Software Engineering, Data Structures and Algorithms, Artificial Intelligence, 3D Computer Vision, Operations Research, Simulation, Statistics, Linear Algebra.

WORK EXPERIENCE

Warehouse AI & Operations Research Engineer (Part-Time), STMicroelectronics (NUS Capstone) **Aug 2026 – Present**

- Architected a layout-agnostic AI put-away engine to run in any STMicroelectronics warehouse worldwide, ingesting each site's shelf coordinates, dimensions, and operating rules so one model adapts to any floor plan.
- Designed a 4-stage shelf-ranking pipeline: movement profiling from pick frequency, dwell time, and demand; filtering on packaging zone, fit, capacity, and restricted areas; Golden Region priority for fast movers; and logged rationale per decision.
- Defined a historical-replay evaluation comparing 3 metrics against current allocation (travel distance per order, handling time, slow-mover Golden Region occupancy), piloting on 2 AMK warehouses: dice and finished goods.

Global Operational AI Systems & Data Engineer (Part-Time), Abbott Laboratories (Contract) **Jun 2026 – Present**

- Develop a chat-based spray-dryer fault-detection tool validated across 5+ plants worldwide, cutting critical-failure diagnosis from days or months of manual raw-signal analysis to within an hour, flagging inconclusive signals rather than false positives.
- Deliver fast metrics for high-stakes decisions such as new product introductions (NPI) and planned stoppages, recommending optimal setpoints from an estimated response surface model and answering engineers' and managers' questions, with rollout planned on Abbott servers for 24/7 access.
- Assemble a 5-stage offline process-intelligence platform for globally distributed spray-drying plants: GPU OCR and layout parsing, recursive chunking, ONNX embeddings, a Chroma vector store, and LangChain over a locally hosted Gemma model.
- Convert global plants' Advanced Process Control models from Excel master files into a single HiGHS linear program returning setpoints, binding constraints, and shadow prices at millisecond latency.
- Build MCP servers, 20+ regression tests, a headless notebook runner, and a deterministic model stand-in for GPU-free CI, with golden snapshots committed for reproducible runs.

Data Analytics & Product Management Intern, Abbott Laboratories (Converted to Contract) **Jan 2026 – Jun 2026**

- Developed a SimPy discrete-event digital twin and OR-Tools CP-SAT solver for hybrid flow-shop scheduling, then compared 3 approaches (heuristic, MIP, and genetic algorithm) to turn production objectives and constraints into decision-ready schedules.
- Validated a non-destructive, 15-stage changeover-data pipeline against 5 years of unseen raw data with zero execution failures across operator- and machine-generated anomalies.
- Deployed a team-built APC simulator to production for live internal users on Azure App Service, with Docker packaging, access restrictions, and runtime and session hardening.
- Trained regional engineering teams across Asia and Europe on repeatable AI workflows spanning 7 areas: prompting, model selection, benchmarks, cost awareness, agents, validation, and safeguards.
- Strengthened Excel, VBA, Python, master-data, and scheduling workflows with 3 safeguards: validation rules, correction tables, and reproducible intermediate stages, leaving every transformation auditable.
- Documented deployment, access provisioning, and technical handover across 3 procedures, so internal manufacturing and engineer-training users stayed supported after transfer to owning teams.

Software Engineer (Full Stack & Cloud), Pro bono, People's Association **Sep 2025 – Sep 2026**

- Volunteered through PA's Sparks Citizen Developer programme to design, build and deploy Churp, a garden-plot balloting platform now live in Tengah with national rollout planned. Project funded by a S$20K Sparks Community Innovation Fund award.
- Led most software development and deployment across admin and public Next.js frontends, a NestJS backend, and GIS mapping.
- Integrated Singpass Login (OIDC) to secure resident sign-in and Myinfo to verify identity and address, keeping garden-plot balloting fair for 150 residents signed up at release.
- Provisioned 10 AWS services with Terraform (ECS Fargate, Multi-AZ RDS Postgres with PostGIS, ALB behind WAFv2, Route 53, ACM, KMS, Secrets Manager, and a Rust auth gateway on Lambda and API Gateway), shipped via keyless GitHub Actions CI/CD on IAM OIDC across 5 repos with gated migrations, RDS snapshots, and auto-rollback.
- Led deployment planning, multi-repo coordination, and stakeholder management in an 11-person team across People's Association, Tengah community partners, and PA ICT (GovTech).
- Surveyed garden sites into geospatial datasets for mapping, plot allocation, and site planning, and designed rules-based balloting across 4 control points (eligibility, conflict resolution, exceptions, admin oversight), keeping allocation auditable.

RESEARCH EXPERIENCE

Research Intern, Digital Supply Chain Group (Incoming), A*STAR **Jan 2027 – May 2027**

- Selected for a 4-month research attachment in ARTC's Digital Supply Chain Group, running alongside final-semester studies.

Computer Vision Researcher, BlendED x Amazon Robotics (AI+X Global Challenge) **Jul 2026 – Oct 2026**

- Tackled an Amazon Robotics brief spanning 3 workstreams (camera ISP, super-resolution, image restoration) to make low-cost cameras rival far pricier sensors, mentored by Dan Shao (CTO, Zelig; former Principal Computer Vision Engineer, iRobot).

- Reproduced a published UNet super-resolution model in PyTorch from broken author code, recovering its undocumented evaluation protocol and training 12 models across 3 scales, 2 datasets, and 2 losses, matching reported results at 8x and landing within 0.78 dB at 2x and 4x.
- Diagnosed a robustness gap where fixed-blur models lost most PSNR gains under unseen blur, then added randomised-blur training and fine-tuning to recover up to 0.7 dB across 3 real-world blur types: lens, out-of-focus, and resampling.
- Authored and presented a super-resolution literature review spanning multi-image, video, and single-image methods, organised around camera information loss and recovery through calibration, extra frames, or learned priors, comparing 4 model families on fidelity and hallucination risk.
- Deployed 3 trained models as a public web app: a one-click Flask inference API on a Colab T4 GPU loading weights from a GitHub release, a secure tunnel on a reserved domain, and a hosted front end for image upload, scale selection, and download.

PROJECTS

- Team Lead & Finalist, Swarmline: SG Defence Tech Hackathon (1,300+ applicants, Sep 2026): Directed a 5-person team building decentralised drone-swarm coordination in a deterministic Unreal Engine 5.8 C++ simulation; under a ground-link jam, 30 drones confirmed all 8 walking targets in 10 of 10 seeded runs, versus 2.1 of 8 on average for an operator-run fleet.
- Personal Project, Hailo Edge AI Training and Deployment (Mar 2026): Trained YOLO26 through transfer learning and reshaped its architecture with the Hailo Dataflow Compiler, running document detection on a Hailo-8L rated 13 TOPS at 1.5 W to speed up large-batch scanning.
- Finalist, Volt Pulse SG: SMU Hack For Cities (Top 8 Finalist of 50+ teams, Jan 2026): Integrated SEALION embeddings, Supabase vector search, RRF routing, and scheduler-agent planning into an autonomous loop managing electricity and utility savings across multiple HDB households at once.
- Hackathon Project, Waaah Comics: 24-Hour Gesture-to-Comic Build (Jan 2026): Turned MediaPipe 3D hand landmarks into generated comic strips in a 24-hour sprint with Gemini 2.0 Flash, Veo 3, and Konva.js canvas tooling.
- Team Lead, Maritime Deficiency Severity Forecasting: Maritime Hackathon (Jun 2025): Led model training for a maritime-inspection NLP system, fine-tuning a 109M-parameter BERT with DNN layers and gradient clipping on the ASPIRE 2A supercomputer.
- Competitor, DSTA BrainHack 2025 CTF (Feb 2025): Solved web, crypto, reverse-engineering, and systems-security challenges under time pressure, ranking top 25% of 400+ teams.

CO-CURRICULAR ACTIVITIES AND VOLUNTEER EXPERIENCES

Independent Bug Bounty Researcher, YesWeHack **May 2024 – Present**

- Assessed GovTech GBBP12/13, Land Transport Authority, and 13+ other programs for SSRF, CSRF, injection, and access-control flaws, automating recon and disclosure evidence with Burp Suite and Python fuzzing scripts.

Elderly Care Volunteer, NTUC Health Active Ageing Centre **Sep 2025 – Sep 2026**

- Coordinated weekly senior-engagement activities, including home visits to check health, hear life stories, and help with everyday tasks, building empathy for accessible, community-focused product design.

NATIONAL SERVICE

Republic of Singapore Navy, Naval Chef & Base Support Assistant **Jan 2021 – Jan 2023**

- Compressed a month-long manual analysis of 6 months of food-waste data into a 3-minute Excel VBA reporting workflow inside a security-constrained environment.

SKILLS AND CERTIFICATIONS

Languages: Python, TypeScript, JavaScript, Java, Rust, C++, SQL, Bash, Excel VBA
Web and Cloud: React, Next.js, Three.js, NestJS, FastAPI, Flask, Docker, AWS (ECS Fargate, RDS, ALB, WAFv2, Lambda, API Gateway, Route 53, ACM, KMS, Secrets Manager, IAM OIDC), Terraform, AWS SAM, Azure App Service, GitHub Actions and Bitbucket CI/CD, Supabase, Singpass Login and Myinfo (OIDC), API design, GIS mapping
AI and Machine Learning: RAG, AI agents, vector search, LangChain, PyTorch, ONNX, Chroma, BERT/transformer fine-tuning, model evaluation, deep RL (PPO, A2C, DDPG, DQN), GPU training (CUDA) on ASPIRE 2A HPC
Simulation and Operations Research: SimPy, OR-Tools CP-SAT, HiGHS linear and quadratic programming, discrete-event simulation, hybrid flow-shop scheduling
Security: OWASP Top 10, Burp Suite, Wireshark, OAuth2/JWT, SSRF/CSRF/injection testing, fuzzing, secure design
Theory: multi-view geometry (camera calibration, epipolar geometry, PnP, structure-from-motion, bundle adjustment), Bayesian networks, queuing theory, statistical inference (hypothesis testing, regression, bootstrap)
Certifications: NVIDIA Disaster Risk Monitoring with Satellite Imagery